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Radiation Environment for the Low Earth Orbit Programs Division Near Earth Orbit Network Program Assets



Radiation Environment for the Low Earth Orbit Programs Division Near Earth Orbit Network Program Assets Review/Signature/Approval Page

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Preface

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I. Introduction

The purpose of this document is to define the radiation environment for the evaluation of degradation due to total ionizing dose, non-ionizing dose, and single event effects for the Low Earth Orbit Programs Division (LPD) Near Earth Orbit Network (NEON) program assets. The analysis is for a polar sun-synchronous low earth orbit (LEO). The radiation analysis considers a 62-month (5 years and 2 months) design life for a circular orbit with an altitude of 824 km and a 98.705-degree angle of inclination. This analysis includes results at the 50% and 95% confidence levels. The radiation assessment envelopes all LPD program missions during the period from 2028-2043.

II. Radiation Environment

The natural space radiation environment of concern is classified into two populations, 1) the transient particles which include protons and heavier ions of all the naturally occurring elements of the periodic table, and 2) the particles trapped in the Earth's magnetic field, which most importantly includes protons and electrons. The trapped protons have energies up to 100s of MeV and the trapped electrons have energies up to about 10 MeV. The transient radiation consists of galactic cosmic ray particles and particles from solar events (coronal mass ejections and flares). The galactic cosmic rays have low-level fluxes that peak as a function of energy around 300 MeV to 1 GeV per nucleon. The solar eruptions periodically produce energetic protons, alpha particles, heavy ions, and electrons. These particle fluxes decrease monotonically with energy beyond about 1 MeV per nucleon and can reach energies as high as the GeV per nucleon range. All particle fluxes are isotropic and omnidirectional to the first order.

Space also contains low energy plasma of electrons and protons with fluxes up to $10^{12} \text{ cm}^{-2}\text{sec}^{-1}$. The plasmasphere environment and the low energy ($< 0.1 \text{ MeV}$) component of the charged particles are a concern in the near-earth environment. In the outer regions of the magnetosphere and in interplanetary space, the plasma is associated with the solar wind. Because of its low energy, thin layers of material easily stop the plasma, so it is not a hazard to most spacecraft electronics. However, it is damaging to surface materials and differentials in the plasma environment can contribute to spacecraft surface charging and discharging problems [1,2].

III. Description of Radiation Effects

Radiation effects that are important to consider for instrument and spacecraft design fall roughly into three categories: degradation from total ionizing dose (TID), degradation from non-ionizing energy loss (NIEL), and single event effects (SEE). Total ionizing dose in electronics is a cumulative, long-term degradation mechanism due to ionizing radiation—mainly primary protons and electrons and secondary particles arising from interactions between these primary particles and spacecraft materials. It causes threshold shifts, leakage current and timing skews. The effect first appears as parametric degradation of the device and ultimately results in functional failure. It is possible to reduce TID with shielding material that absorbs most electrons and lower energy protons. As shielding is increased, shielding effectiveness decreases because of the difficulty in slowing down the higher energy protons. When manufacturers advertise a part as “rad-hard”, they are almost always referring to its total ionizing dose characteristics. Rad-hard does not usually imply that the part is hard to non-ionizing dose or single event effects. In some cases, a “rad-hard” part may perform significantly worse in the space radiation environment than in the test environment (e.g. Enhanced Low Dose Rate Sensitivity in linear bipolar devices.)

Displacement damage is cumulative, long-term non-ionizing damage due to protons, electrons, and neutrons. These particles produce defects in semiconductor and optical materials that result in material property changes such as carrier

lifetime shortening, mobility degradation and optical transmission. Displacement damage affects the performance of optocouplers (often a component in power devices), solar cells, CCDs, and linear bipolar devices. The effectiveness of shielding depends on the location of the device. For example, cover glass over solar cells reduce electron damage and proton damage by absorbing the low energy particles. Increasing shielding beyond a certain point, however, is not usually effective for optoelectronic components because the high-energy protons penetrate the most feasible spacecraft electronic enclosures. For detectors in instruments, it is necessary to understand the instrument technology and geometry to determine the vulnerability to the environment.

Single event effects (SEE) result from ionization by a single charged particle as it passes through a sensitive region of an electronic or photonic device. SEE can be caused by heavier ions, but for some devices, protons, and neutrons* can also contribute. SEEs are caused either by direct ionization or by secondary particles produced by the interaction of the incident particle with a nucleus in the device material. Incident heavy ions produce SEEs most often by direct ionization. However, in some cases secondary products can cause SEE. Conversely, incident protons most often produce SEEs via secondary products but in some cases, they can be caused by direct ionization. In all instances for incident neutrons, SEEs result from secondary particles produced when the neutron interacts with a nucleus in the device material. Some single event effects are non-destructive, as in the case of single event upsets (SEUs), single event transients (SETs), multiple bit upset (MBUs), single event hard errors (SEEs), etc. Single event effects can also be destructive as in the case of single event latchups (SELs), single event gate ruptures (SEGRs), and single event burnouts (SEBs). The severity of the effect can range from noisy data to system shutdown, depending on the type of effect and the criticality of the system in which it occurs. Shielding is generally not an effective mitigation for single event effects, which can be induced by very penetrating high-energy particles. The preferred method for dealing with destructive failures is to use SEE-hard parts. When SEE-hard parts are not available, latchup protection circuitry is sometimes used in conjunction with failure mode analysis. (Note: Care is necessary when using SEL protection circuitry, because SEL may damage a microcircuit and reduce its reliability even when it does not cause outright failure.) For nondestructive effects, mitigation takes the form of error-detection and correction codes (EDACs), filtering circuitry, etc.

Total ionizing dose is primarily caused by protons and electrons trapped in the Van Allen belts and solar event protons. As electrons are slowed down, their interactions with orbital electrons of the shielding material produce a secondary photon radiation known as bremsstrahlung. Generally, the dose due to galactic cosmic ray ions and proton secondaries and bremsstrahlung is small compared to other sources in modern spacecraft having typical shielding. For surface degradation, it is also important to include the effects of very low energy particles.

SEE can be induced by heavy ions (solar events and galactic cosmic rays) and, in some devices, protons (trapped and solar events) and neutrons*. Displacement damage is primarily due to trapped and solar protons and neutrons*. High-energy electrons can also contribute to displacement damage, especially for lightly shielded applications. Spacecraft charging can occur on the surface of the spacecraft due to low energy electrons. Deep dielectric charging occurs when high-energy electrons penetrate the spacecraft and collect in dielectric materials.

*Neutrons in space are produced by interactions of primary particles with spacecraft materials and planet soils. In avionics applications it is also necessary to consider neutrons that are produced by interactions of primary particles with the atmosphere. energy electrons. Deep dielectric charging occurs when high-energy electrons penetrate the spacecraft and collect in dielectric materials.

IV. NEON Program Mission

The mission orbit is planned to be circular at an altitude of 824 km, and 98.705-degree angle of inclination operating in either the 13:30 or 17:30 LTAN (Local Time of Ascending Node). Current planning is for a 5.17-year mission and launch date between 2028-2043.

The NEON program orbit radiation environment will consist of protons and electrons trapped in the Van Allen belts, of protons and heavier ions from solar events, galactic cosmic ray ions, and solar wind plasma consisting of low energy electrons, protons, and heavier ions.

V. Total Dose and Degradation

The total ionizing dose accumulation causes performance degradation and failure on electronic and photonic components. Non-ionizing energy loss in materials (atomic displacement damage) causes degradation of systems consisting of components such as solar cells, optoelectronics, and detectors. The low energy particles also contribute to the erosion of surfaces.

A. Degradation Environment

1. The Plasma Environment

At the NEON orbits, low energy particles from the solar wind plasma inside and outside the magnetotail contribute to the degradation of surface materials and cause charging effects. These charging effects should be considered in the spacecraft design [3,4]. Daily values of trapped electrons can fluctuate by two to three orders of magnitude depending on the level of activity on the sun and within the magnetosphere.

2. High Energy Particles – Spacecraft Incident Fluences

The spacecraft incident particle fluence levels given in this document can be used for standard solar cell analyses that consider the cover glass thickness of the cell. There are three types of high energy particles that contribute significantly to solar cell degradation: trapped protons and trapped electrons encountered in the Van Allen belts and protons from solar events that are most likely to occur during solar maximum years. These fluence levels can also be used for other analyses. Most analysis methods require that the surface incident particles be transported through the materials surrounding the sensitive components.

The trapped particle fluxes including plasma were estimated with the new AP-9 and AE-9 models at the 50% (median) and 95% confidence levels [5]. It is recommended that the 95% confidence level be used for providing design margin.

Table A1 gives the incident trapped proton fluence. **Figure 1** is a plot of the fluence-energy trapped proton spectrum. The ordinate shows the proton fluence for the 5.17-year mission having energy, E , greater than the corresponding value shown on the abscissa.

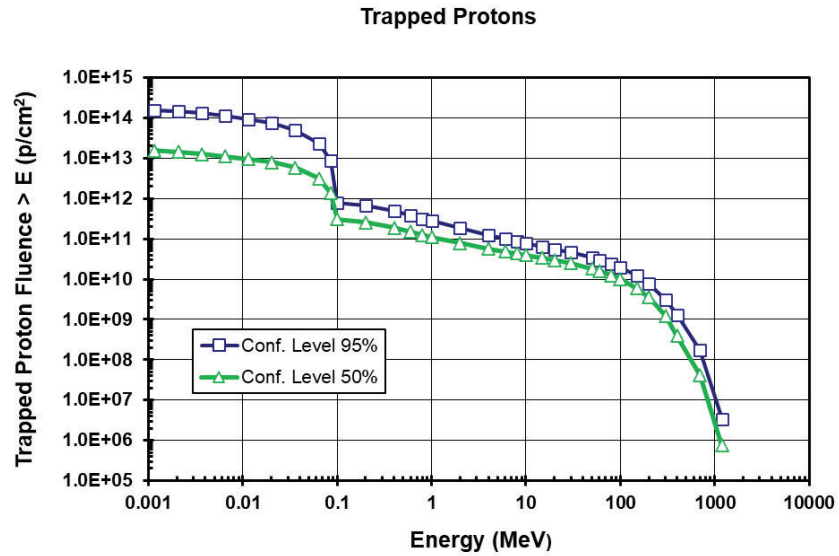


Figure 1: Surface incident integral trapped proton fluences for the 5.17-year NEON program satellite mission(s) at the 50% and 95% confidence levels.

Table A2 gives the incident trapped electron fluences. **Figure 2** is a plot of the fluence-energy trapped electron spectra.

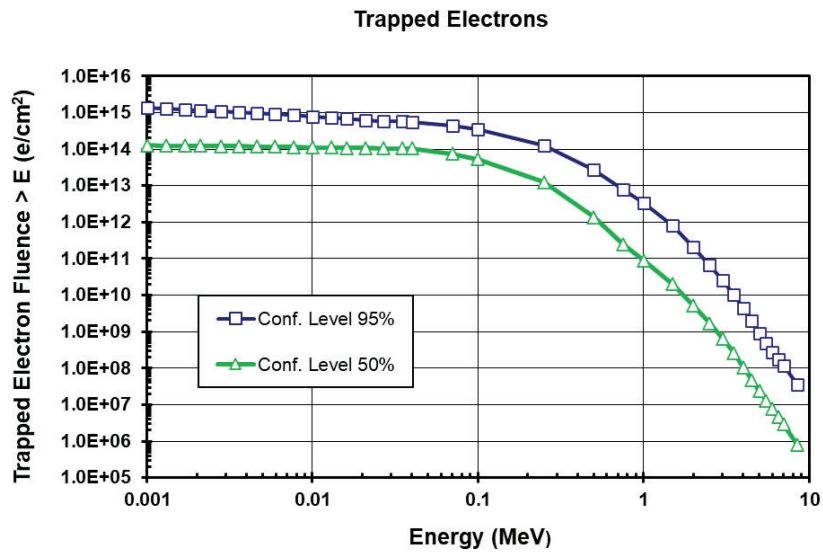


Figure 2: Surface incident integral trapped electron fluences for the 5.17-year NEON program satellite mission(s) at the 50% and 95% confidence levels.

The solar proton levels were estimated using the Emission of Solar Protons (ESP) model [6]. The ESP model is based on satellite data from solar cycles 20 - 23. A worst-case assumption was made that the NEON program would occur entirely during the solar maximum phase of the cycle. The distribution of the fluences for the events is obtained from maximum entropy theory, and design limits in the worst-case models are obtained from extreme value theory.

Solar particle events are often described with probability distributions because of the statistical nature of the event magnitudes and times of occurrence. **Table A3** and **Figure 3** give the incident solar proton fluence-energy spectra for the 50% and 95% confidence levels for consistency with the results for trapped particles. Predictions that are based on probability distributions do not increase linearly with time. Only the mean value of the distribution increases linearly. Therefore, these estimates at high confidence levels lose accuracy if extrapolated to longer or shorter mission durations.

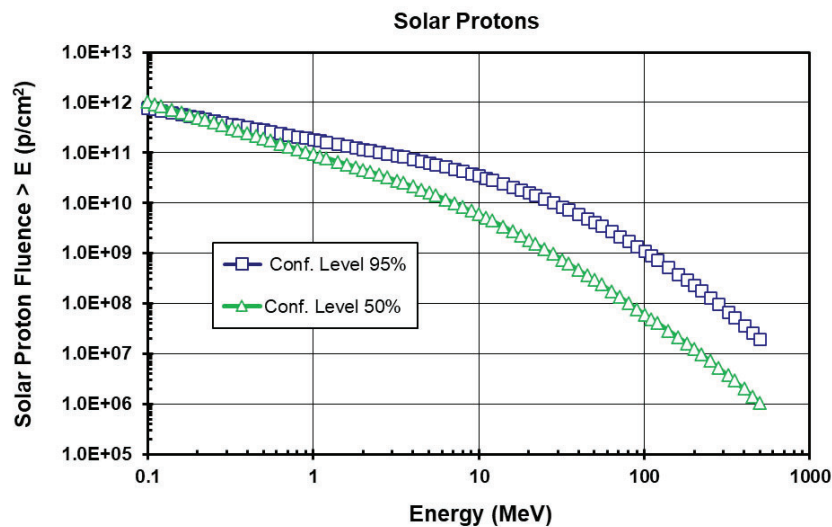


Figure 3: Surface incident integral solar proton fluences for the 5.17-year NEON program satellite mission(s) at the 50% and 95% confidence levels.

B. Total Ionizing Dose Estimates

1. Top Level Ionizing Dose Requirement

Doses are calculated from the surface incident integral fluences as a function of aluminum shield thickness for a simple geometry. The geometry model used for spacecraft applications is often the solid sphere. The solid sphere doses represent an upper boundary for the dose inside an actual spacecraft and are used as a top-level requirement. In cases where the amount of shielding surrounding a sensitive location is difficult to estimate, a more detailed analysis of the

geometry of the spacecraft structure may be necessary to evaluate the expected dose levels. This is done modeling the electronic boxes or instruments and the spacecraft structure. The amount of shielding surrounding selected sensitive locations is estimated using solid angle sectoring and 3-dimensional ray tracing. This is then combined with full Monte Carlo transport simulations of particle trajectories through the structure to evaluate the dose and is described in more detail in section 2.

Tables A4 and **A5** give the expected total ionizing doses for NEON program satellite mission(s) at the 50% and 95% confidence levels, respectively. **Figures 4** and **5** show the results, which include all the components and the total. They are calculated as a function of aluminum shield thickness in units of krad in silicon. For a nominal 100 mils (2.54 mm) of equivalent aluminum shielding and the 5.17-year mission, the expected dose at the 95% confidence level is 35.4 krad-Si. This is the top-level requirement.

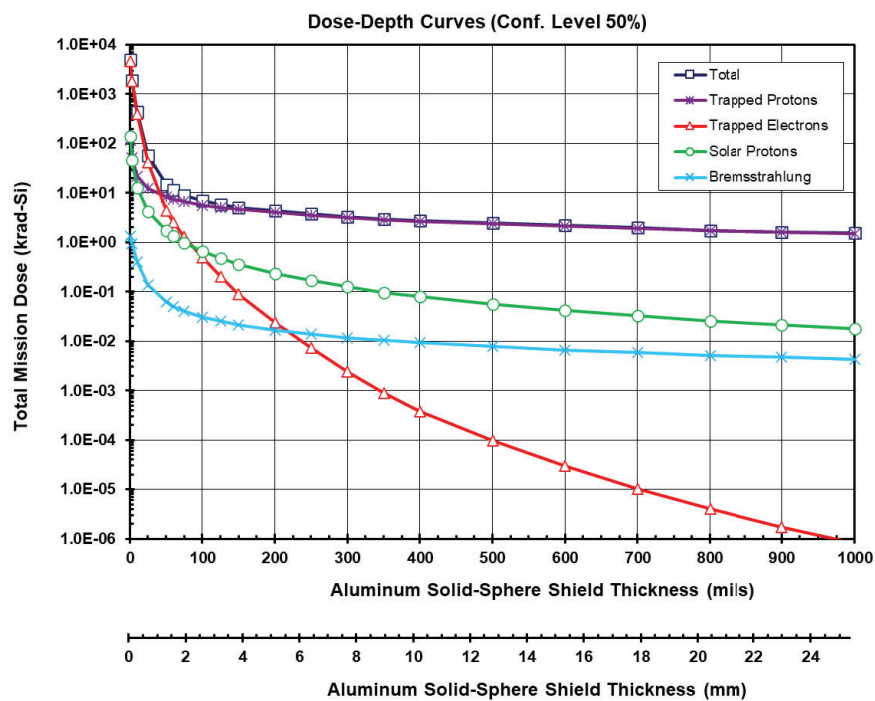


Figure 4: NEON program satellite mission(s) total ionizing dose curves at the 50% confidence level.

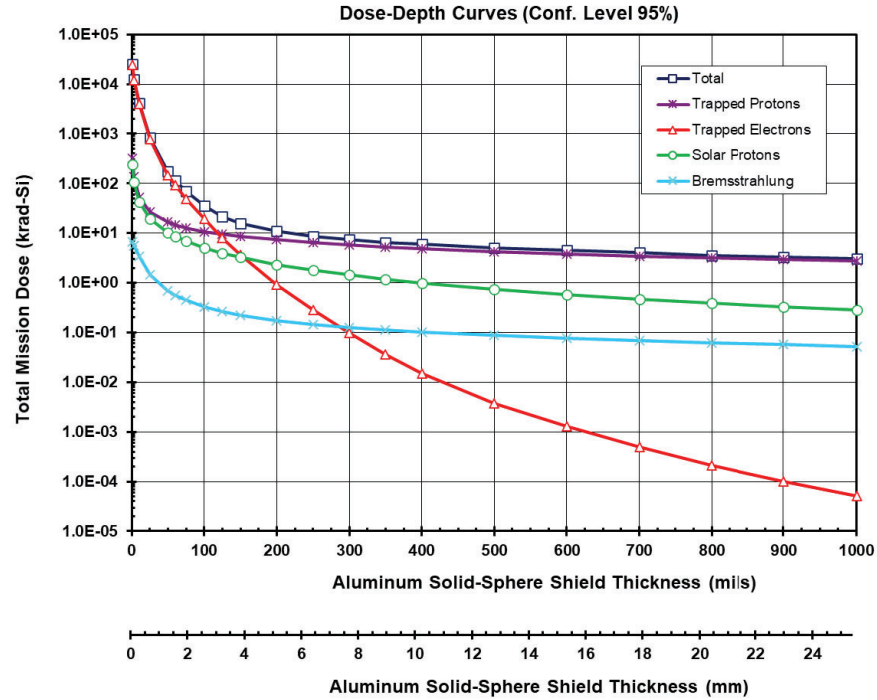


Figure 5: NEON program satellite mission(s) total ionizing dose curves at the 95% confidence level.

2. Dose at Specific Spacecraft Locations

In cases where parts cannot meet the top-level design requirement and a “harder” part cannot be substituted, it is often beneficial to employ more accurate methods of determining the dose exposure for some spacecraft components to qualify the parts. One such method for calculating total dose, solid angle sectoring/3-dimensional ray tracing, and radiation transport simulation, is accomplished in three steps:

- 1) Model the spacecraft structure:
 - develop a 3-D model of the spacecraft structures and components
 - develop a material library
 - define sensitive locations
- 2) Model the radiation environment:
 - define the spacecraft incident radiation environment, as in this document
 - develop an attenuation model using the shielding configurations (similar to dose-depth curves).
- 3) Obtain results for each sensitive location:
 - divide the structural model into solid angle sectors
 - ray trace through the sectors to calculate the material mass distribution
 - use the ray trace results to simulate total doses from a Monte Carlo transport calculation.

Once the basic structural model has been defined, total doses can be obtained for any location in the spacecraft in a relatively short time. The value of dose mitigation measures can then be evaluated by adding the changes to the model

and recalculating the total dose. This may help avoid unnecessary use of expensive or increasingly unavailable radiation hardened parts. Simplified models analogous to this may also be implemented such as modeling only an electronic box or only an instrument.

C. Displacement Damage Dose Estimates

The displacement damage dose or non-ionizing dose is the mean energy deposited in a material that goes into atomic displacements divided by the mass of the material. It is analogous to the ionizing dose except that the energy considered produces displacements in a semiconductor lattice.

1. Top Level Displacement Damage Requirement

Starting with the surface incident integral fluences, the displacement damage dose (DDD) can be calculated as a function of aluminum shield thickness using solid sphere geometry. It can be written

$$DDD = C \int NIEL(E) \times \phi(E) dE$$

where NIEL(E) is the Non-Ionizing Energy Loss as a function of proton energy [7], $\phi(E)$ is the proton fluence as a function of energy and C is a constant. If C is taken as unity then DDD is in units of MeV/g. Results are material dependent and are often converted to an equivalent fluence of particles according to Non-Ionizing Energy Loss (NIEL) [7]. For silicon devices equivalent fluences of 1, 10, 50 and 63 MeV protons and 1 MeV neutrons are given in **Tables A6** and **A7** as a function of shield thickness and for the 50% and 95% confidence levels, respectively, while **Figure 6** and **Figure 7** shows key results. For example, if 100 mils of Al shielding are assumed the displacement damage requirement for a silicon device in terms of 10 MeV equivalent proton fluence at the 95% confidence level is $5.741 \times 10^{10} \text{ cm}^{-2}$. For gallium arsenide devices the results are given in **Table A8**, **Table A9**, **Figure 8** and **Figure 9**.

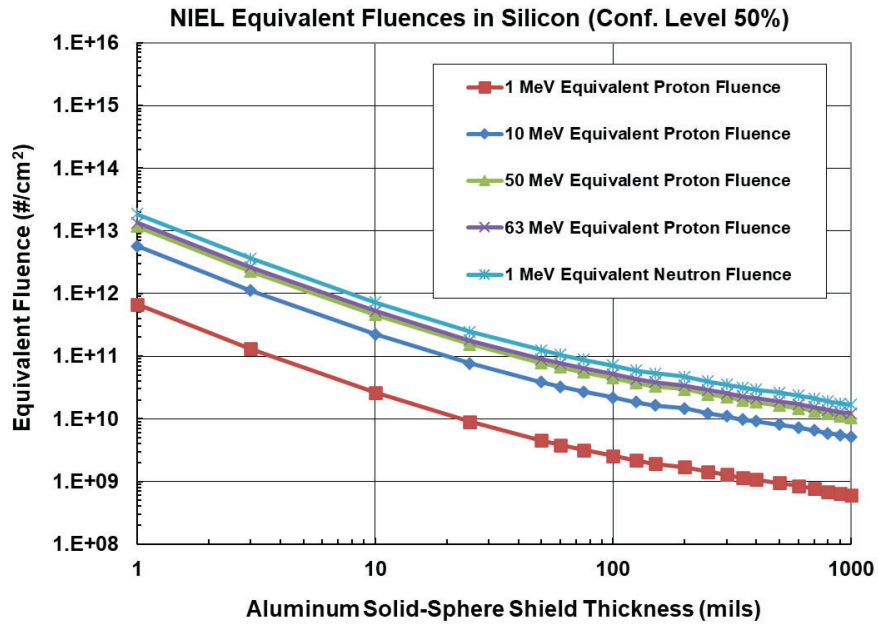


Figure 6: NIEL Equivalent Proton and Neutron Fluences in Silicon at the Center of Solid Aluminum Spheres. Values for the 50% confidence level are shown.

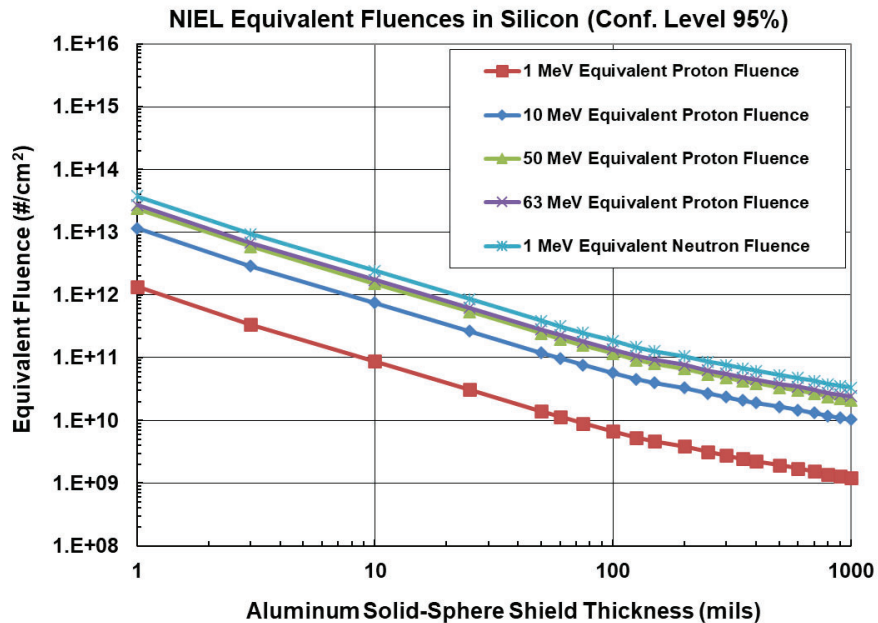


Figure 7: NIEL Equivalent Proton and Neutron Fluences in Silicon at the Center of Solid Aluminum Spheres. Values for the 95% confidence level are shown.

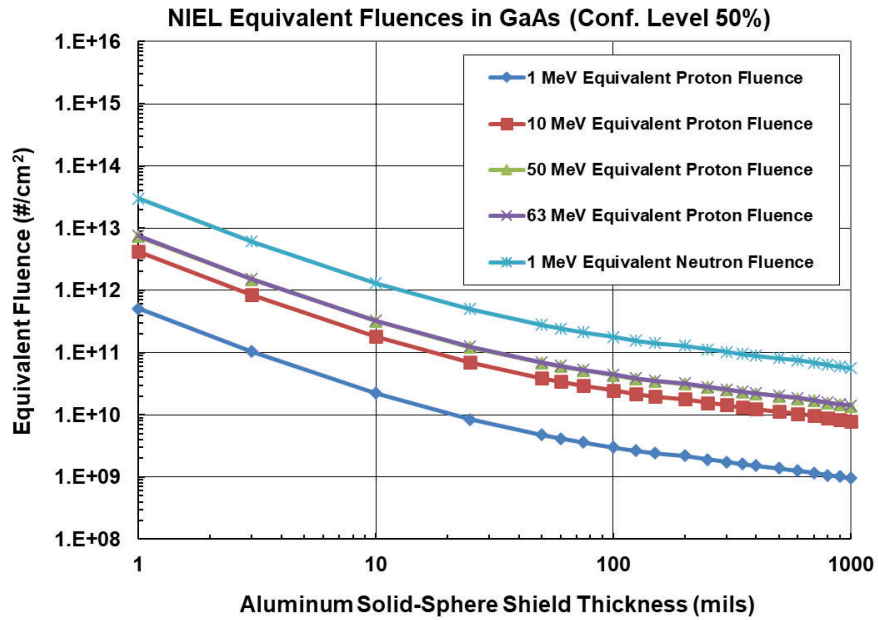


Figure 8: NIEL Equivalent Proton and Neutron Fluences in Gallium Arsenide at the Center of Solid Aluminum Spheres. Values for the 50% confidence level are shown.

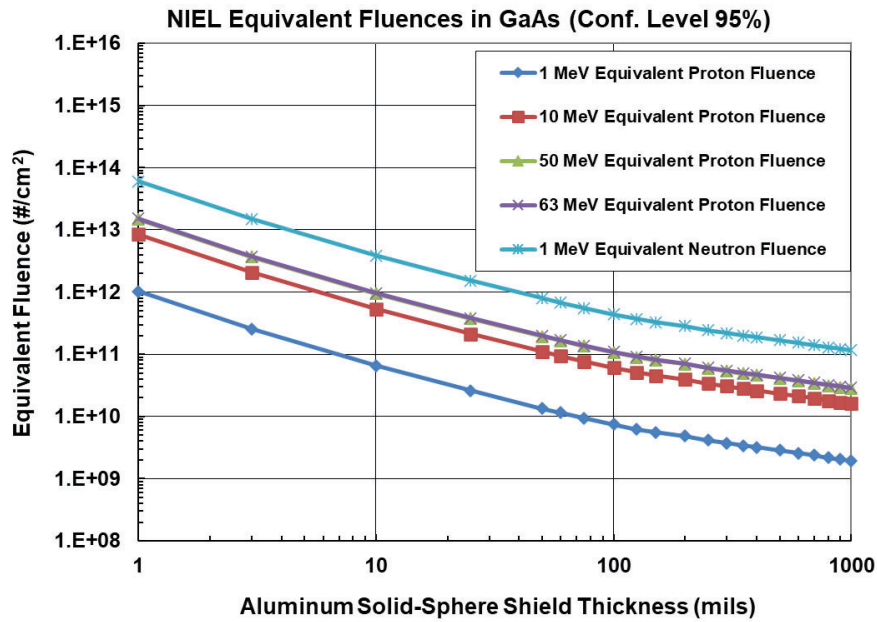


Figure 9: NIEL Equivalent Proton and Neutron Fluences in Gallium Arsenide at the Center of Solid Aluminum Spheres. Values for the 95% confidence level are shown.

VI. Single Event Effects Analysis

A. Heavy Ion Induced Single Event Effects

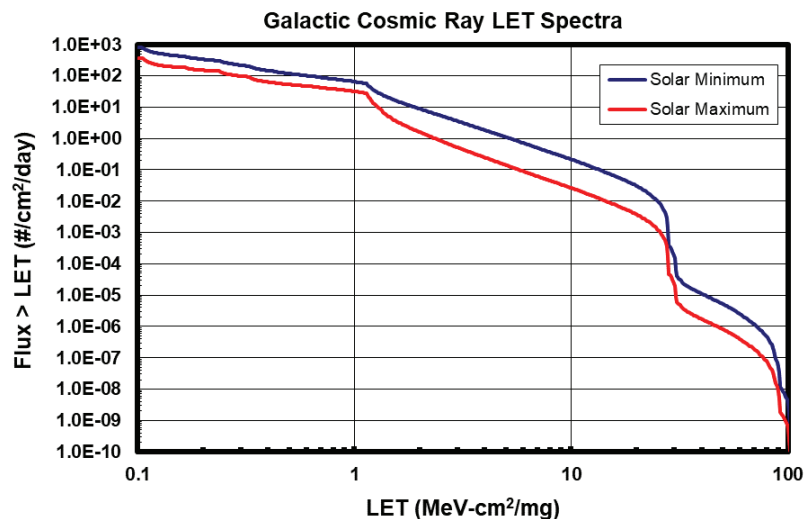
Some electronic devices are susceptible to SEE, e.g., single event upsets, single event latch-up, single event burn-out. Heavy ions such as galactic cosmic rays (GCR) and solar heavy ions most often cause SEEs by the direct deposit of charge. The metric traditionally used to describe this is the ion's linear energy transfer (LET). The LET is the energy lost by the ion per unit path length in the material of interest. To parameterize SEEs in terms of LET, the heavy ion abundances and energy distributions in the environment are converted to LET spectra. Once specific parts are selected for the mission and, if necessary, characterized by laboratory testing, the LET spectra for the heavy ions in the space radiation environment are integrated with the device characterization to estimate SEE rates.

It should be noted that the LET metric may not be valid for some modern devices. This can be because of their highly scaled nature, the complexity of the geometry, the increased use of metal over-layers or other reasons not yet understood. Such instances underscore the importance of SEE testing and may require a more sophisticated SEE rate calculation than described above.

Geomagnetic attenuation, i.e., attenuation by the Earth's magnetic field of galactic cosmic ray and solar particle fluxes for the NEON program satellite mission(s) orbit is accounted for in this document.

1. Galactic Cosmic Rays

The GCR fluxes for elements hydrogen through uranium were used to calculate daily LET spectra for a nominal shielding thickness of 100 mils aluminum. Results are given in **Table A10** and **Figure 10**. The range of the cosmic ray abundances is bounded by the extrema of the solar maximum and minimum phases of the solar cycle with the highest values occurring during the solar minimum phase. The LET spectra values are given for the highest and lowest point of the solar cycle. The CREME96 [8] model was used to obtain the GCR abundances. This model has an accuracy of 25-40%.



**Figure 10: Integral LET spectra for galactic cosmic ray ions hydrogen through uranium
assuming 100 mils of aluminum shielding**

2. Solar Heavy Ions

The heavy ions from coronal mass ejections and solar flares can also produce SEEs. As discussed previously these events are statistical in nature. The solar event fluxes for the elements hydrogen through uranium were used to calculate worst-case LET spectra for 100 mils aluminum shielding. This is shown in **Table A11** and **Figure 11**. The intensity during an event is shown for 2 worst case situations – the flux averaged over the peak 5 minutes and the flux averaged over the peak 24 hours. The CREME96 model has been used for these situations, which assumes that the well-known October 1989 event is the worst case.

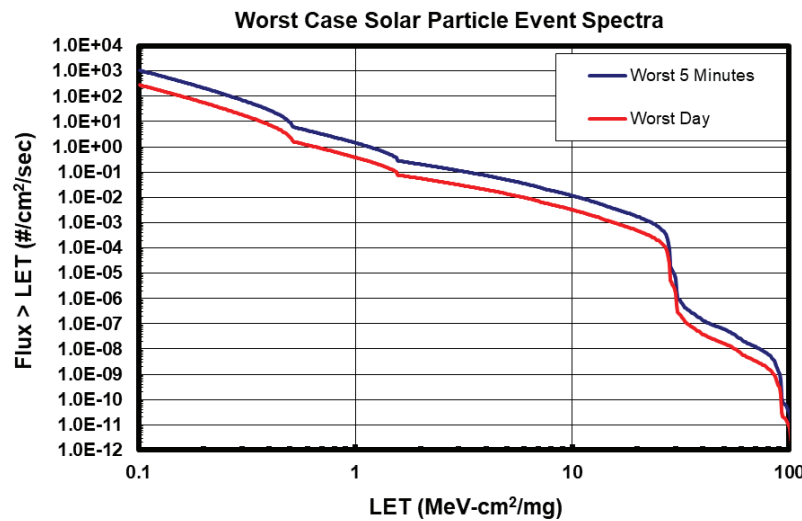


Figure 11: Integral LET spectra for hydrogen through uranium for 2 worst case situations: (1) average over the peak 5 minutes of the October 1989 event, (2) average over the worst day during the October 1989 event.

B. Proton Induced Single Event Effects

In some devices, SEEs are also induced by protons. In most cases, protons do not generate enough ionization to produce the charge necessary for SEEs to occur. More typically, protons cause SEEs through secondary particles via nuclear interactions, that is, spallation and fractionation products. The proton energy (more than the LET) is important in the production of the secondary particles that cause the SEEs, so device sensitivity to these particles is typically expressed as a function of proton energy.

1. Worst-Case Proton Fluxes

For the NEON program satellite mission(s) orbit the worst-case proton fluxes are from solar protons. These can reach levels that are a concern for short-term SEE rates. Worst-case solar proton fluxes for a nominal 100 mils of aluminum shielding are given in **Table A12** and shown in **Figure 12**. These values are averaged over the peak 5 minutes during

the event. This was calculated at the 95% and 50% confidence levels using the ESP model. These results can be used to calculate worst-case SEE rates caused by protons.

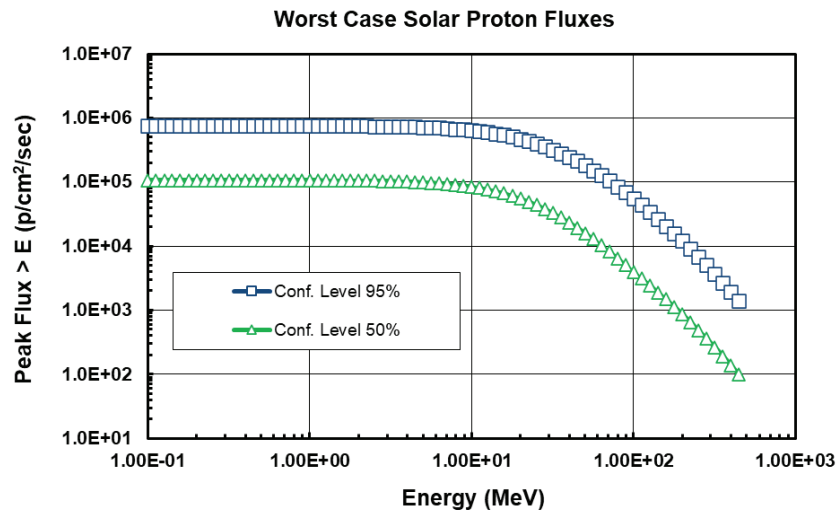


Figure 12: Worst case integral proton fluxes for single event effects evaluation.

This results from solar protons and is averaged over a 5-minute interval at the 95% confidence level.

Calculations are done for 100 mils of aluminum shielding.

VII. Spacecraft Charging and Discharging

Surface charging and deep dielectric charging must also be evaluated for the NEON program satellite mission(s)[4]. Both are potential problems during passes through the Van Allen belts. During a pass through the belts, high levels of charge can build-up on spacecraft surfaces (mainly due to low energy electrons) and in dielectrics in the interior (mainly due to high energy electrons).

VIII. Summary

A top-level worst-case radiation environment specification was presented for the NEON program satellite mission(s). The total dose environment is moderate. However, some of the newer technologies and commercial-off-the-shelf (COTS) devices are very soft to total ionizing dose and displacement damage effects. The single event effects environment is somewhat more severe. Both total dose and single event effects require careful radiation testing, part selection and mitigation schemes.

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Appendices

Table A1
Surface Incident Integral Trapped Proton Fluences for
the 5.17-year NEON Program Satellite Mission(s)

Energy, E	CL 50%	CL 95%
(MeV):	Fluence > E	Fluence > E
(p/cm²):	(p/cm²):	(p/cm²):
0.00115	1.565E+13	1.552E+14
0.0021	1.429E+13	1.452E+14
0.0037	1.276E+13	1.313E+14
0.0065	1.124E+13	1.143E+14
0.01155	9.675E+12	9.510E+13
0.0204	7.980E+12	7.455E+13
0.036	5.921E+12	5.098E+13
0.06375	3.165E+12	2.339E+13
0.085	1.416E+12	8.704E+12
0.1	3.112E+11	8.056E+11
0.2	2.587E+11	6.671E+11
0.4	1.894E+11	4.846E+11
0.6	1.496E+11	3.793E+11
0.8	1.250E+11	3.139E+11
1	1.104E+11	2.749E+11
2	7.899E+10	1.875E+11
4	5.738E+10	1.250E+11
6	4.847E+10	9.981E+10
8	4.333E+10	8.617E+10
10	3.978E+10	7.749E+10
15	3.378E+10	6.381E+10
20	2.984E+10	5.550E+10
30	2.492E+10	4.589E+10
50	1.840E+10	3.401E+10
60	1.610E+10	3.001E+10
80	1.261E+10	2.401E+10
100	1.006E+10	1.955E+10
150	5.940E+09	1.215E+10
200	3.572E+09	7.734E+09
300	1.234E+09	3.115E+09
400	3.880E+08	1.257E+09
700	4.139E+07	1.727E+08
1200	7.442E+05	3.434E+06
2000	0.000E+00	0.000E+00

Table A2
Surface Incident Integral Trapped Electron Fluences for the 5.17-year for
the 5.17-year NEON Program Satellite Mission(s)

Energy, E	CL 50%	CL 95%
(MeV):	Fluence > E	Fluence > E
(e/cm²):	(e/cm²):	(e/cm²):
0.001	1.260E+14	1.329E+15
0.0013	1.249E+14	1.259E+15
0.0017	1.237E+14	1.188E+15
0.0021	1.228E+14	1.137E+15
0.0028	1.215E+14	1.071E+15
0.0036	1.201E+14	1.014E+15
0.0046	1.186E+14	9.616E+14
0.0059	1.167E+14	9.072E+14
0.0077	1.145E+14	8.461E+14
0.01	1.121E+14	7.842E+14
0.013	1.097E+14	7.200E+14
0.016	1.080E+14	6.722E+14
0.021	1.062E+14	6.226E+14
0.027	1.050E+14	5.916E+14
0.035	1.041E+14	5.715E+14
0.04	1.038E+14	5.639E+14
0.07	7.355E+13	4.409E+14
0.1	5.264E+13	3.490E+14
0.25	1.253E+13	1.271E+14
0.5	1.368E+12	2.681E+13
0.75	2.489E+11	7.718E+12
1	9.097E+10	3.313E+12
1.5	2.036E+10	8.017E+11
2	5.176E+09	2.110E+11
2.5	1.690E+09	6.666E+10
3	6.597E+08	2.562E+10
3.5	2.617E+08	1.030E+10
4	1.064E+08	4.340E+09
4.5	4.743E+07	1.954E+09
5	2.364E+07	9.217E+08
5.5	1.301E+07	4.789E+08
6	7.585E+06	2.769E+08
6.5	4.558E+06	1.739E+08
7	2.900E+06	1.170E+08
8.5	7.777E+05	3.591E+07
10	0.000E+00	0.000E+00

Table A3
Surface Incident Integral Solar Proton Fluences for the 5.17-year for
the 5.17-year NEON Program Satellite Mission(s)

Energy, E	CL 50% Fluence > E	CL 95% Fluence > E
(MeV):	(p/cm²):	(p/cm²):
0.1	1.027E+12	7.855E+11
0.11	9.321E+11	7.404E+11
0.12	8.523E+11	7.011E+11
0.14	7.258E+11	6.359E+11
0.16	6.314E+11	5.844E+11
0.18	5.588E+11	5.426E+11
0.2	5.012E+11	5.079E+11
0.22	4.543E+11	4.784E+11
0.25	3.979E+11	4.414E+11
0.28	3.538E+11	4.110E+11
0.32	3.080E+11	3.779E+11
0.35	2.808E+11	3.572E+11
0.4	2.445E+11	3.284E+11
0.45	2.164E+11	3.049E+11
0.5	1.941E+11	2.854E+11
0.55	1.760E+11	2.689E+11
0.63	1.528E+11	2.468E+11
0.71	1.350E+11	2.289E+11
0.8	1.193E+11	2.123E+11
0.9	1.056E+11	1.972E+11
1	9.472E+10	1.846E+11
1.1	8.588E+10	1.739E+11
1.2	7.854E+10	1.647E+11
1.4	6.689E+10	1.494E+11
1.6	5.820E+10	1.373E+11
1.8	5.151E+10	1.275E+11
2	4.621E+10	1.194E+11
2.2	4.190E+10	1.125E+11
2.5	3.671E+10	1.038E+11
2.8	3.259E+10	9.659E+10
3.2	2.806E+10	8.830E+10
3.5	2.519E+10	8.283E+10
4	2.135E+10	7.514E+10
4.5	1.846E+10	6.895E+10
5	1.616E+10	6.374E+10
5.5	1.425E+10	5.909E+10
6.3	1.180E+10	5.280E+10
7.1	9.970E+09	4.769E+10
8	8.369E+09	4.285E+10
9	7.012E+09	3.843E+10
10	5.973E+09	3.477E+10
11	5.135E+09	3.157E+10

Table A3 (continued)
Surface Incident Integral Solar Proton Fluences for the 5.17-year for
the 5.17-year NEON Program Satellite Mission(s)

Energy, E	CL 50%	CL 95%
(MeV):	Fluence > E	Fluence > E
(p/cm²):	(p/cm²):	(p/cm²):
12	4.456E+09	2.880E+10
14	3.441E+09	2.437E+10
16	2.732E+09	2.095E+10
18	2.218E+09	1.822E+10
20	1.835E+09	1.601E+10
22	1.540E+09	1.418E+10
25	1.207E+09	1.194E+10
28	9.683E+08	1.019E+10
32	7.410E+08	8.381E+09
35	6.190E+08	7.331E+09
40	4.696E+08	5.953E+09
45	3.663E+08	4.916E+09
50	2.930E+08	4.125E+09
55	2.390E+08	3.504E+09
63	1.771E+08	2.748E+09
71	1.353E+08	2.199E+09
80	1.028E+08	1.745E+09
90	7.787E+07	1.375E+09
100	6.086E+07	1.104E+09
110	4.932E+07	9.032E+08
120	4.100E+07	7.509E+08
140	2.905E+07	5.321E+08
160	2.131E+07	3.902E+08
180	1.615E+07	2.958E+08
200	1.257E+07	2.302E+08
220	9.977E+06	1.827E+08
250	7.222E+06	1.322E+08
280	5.362E+06	9.820E+07
320	3.728E+06	6.827E+07
350	2.926E+06	5.359E+07
400	2.016E+06	3.692E+07
450	1.439E+06	2.636E+07
500	1.078E+06	1.974E+07

Table A4
NEON Program Satellite Mission(s)
Total Ionizing Dose at the Center of Solid Aluminum Spheres
CL = 50%

Aluminum Shield Thickness			Mission Dose (krad-Si)				
mils	mm	g/cm ²	Total	Trap e	Brem	Trap p	Solar p
1	0.0254	0.00686	4.97E+03	4.71E+03	1.35E+00	1.18E+02	1.39E+02
3	0.0762	0.02057	1.93E+03	1.83E+03	9.31E-01	5.19E+01	4.66E+01
10	0.2540	0.06855	4.32E+02	3.97E+02	4.05E-01	2.17E+01	1.27E+01
25	0.6350	0.17138	5.80E+01	4.11E+01	1.38E-01	1.25E+01	4.23E+00
50	1.2700	0.34276	1.47E+01	4.44E+00	6.18E-02	8.45E+00	1.71E+00
60	1.5240	0.41131	1.16E+01	2.58E+00	5.12E-02	7.61E+00	1.33E+00
75	1.9050	0.51414	9.02E+00	1.29E+00	4.10E-02	6.70E+00	9.81E-01
100	2.5400	0.68552	6.89E+00	4.90E-01	3.11E-02	5.72E+00	6.54E-01
125	3.1750	0.85690	5.77E+00	2.00E-01	2.53E-02	5.07E+00	4.75E-01
150	3.8100	1.02828	5.12E+00	9.03E-02	2.15E-02	4.65E+00	3.63E-01
200	5.0800	1.37104	4.39E+00	2.37E-02	1.66E-02	4.11E+00	2.35E-01
250	6.3500	1.71380	3.73E+00	7.26E-03	1.37E-02	3.55E+00	1.67E-01
300	7.6200	2.05656	3.31E+00	2.40E-03	1.18E-02	3.17E+00	1.27E-01
350	8.8900	2.39932	3.00E+00	8.91E-04	1.04E-02	2.89E+00	9.74E-02
400	10.1600	2.74208	2.79E+00	3.84E-04	9.29E-03	2.70E+00	7.91E-02
500	12.7000	3.42760	2.46E+00	9.61E-05	7.74E-03	2.39E+00	5.61E-02
600	15.2400	4.11312	2.20E+00	2.94E-05	6.58E-03	2.16E+00	4.18E-02
700	17.7800	4.79864	1.97E+00	1.02E-05	5.85E-03	1.93E+00	3.25E-02
800	20.3200	5.48416	1.76E+00	4.00E-06	5.20E-03	1.73E+00	2.56E-02
900	22.8600	6.16969	1.64E+00	1.73E-06	4.69E-03	1.61E+00	2.13E-02
1000	25.4000	6.85521	1.54E+00	8.12E-07	4.30E-03	1.52E+00	1.79E-02

Table A5
NEON Program Satellite Mission(s)
Total Ionizing Dose at the Center of Solid Aluminum Spheres
CL = 95%

Aluminum Shield Thickness			Mission Dose (krad-Si)				
mils	mm	g/cm ²	Total	Trap e	Brem	Trap p	Solar p
1	0.0254	0.00686	2.48E+04	2.42E+04	6.69E+00	3.25E+02	2.37E+02
3	0.0762	0.02057	1.23E+04	1.20E+04	5.58E+00	1.38E+02	1.06E+02
10	0.2540	0.06855	4.13E+03	4.04E+03	3.35E+00	5.15E+01	4.20E+01
25	0.6350	0.17138	8.38E+02	7.91E+02	1.45E+00	2.65E+01	1.97E+01
50	1.2700	0.34276	1.74E+02	1.46E+02	6.80E-01	1.67E+01	1.03E+01
60	1.5240	0.41131	1.16E+02	9.16E+01	5.61E-01	1.48E+01	8.59E+00
75	1.9050	0.51414	6.86E+01	4.85E+01	4.44E-01	1.28E+01	6.81E+00
100	2.5400	0.68552	3.54E+01	1.93E+01	3.32E-01	1.07E+01	5.10E+00
125	3.1750	0.85690	2.17E+01	8.06E+00	2.67E-01	9.39E+00	3.93E+00
150	3.8100	1.02828	1.56E+01	3.65E+00	2.25E-01	8.54E+00	3.24E+00
200	5.0800	1.37104	1.09E+01	9.32E-01	1.75E-01	7.43E+00	2.32E+00
250	6.3500	1.71380	8.60E+00	2.88E-01	1.46E-01	6.40E+00	1.77E+00
300	7.6200	2.05656	7.36E+00	9.70E-02	1.26E-01	5.71E+00	1.43E+00
350	8.8900	2.39932	6.50E+00	3.58E-02	1.13E-01	5.18E+00	1.16E+00
400	10.1600	2.74208	5.92E+00	1.48E-02	1.02E-01	4.82E+00	9.83E-01
500	12.7000	3.42760	5.09E+00	3.74E-03	8.73E-02	4.26E+00	7.40E-01
600	15.2400	4.11312	4.49E+00	1.28E-03	7.58E-02	3.84E+00	5.81E-01
700	17.7800	4.79864	4.00E+00	4.93E-04	6.87E-02	3.46E+00	4.73E-01
800	20.3200	5.48416	3.57E+00	2.12E-04	6.23E-02	3.12E+00	3.88E-01
900	22.8600	6.16969	3.30E+00	1.00E-04	5.70E-02	2.92E+00	3.32E-01
1000	25.4000	6.85521	3.10E+00	5.04E-05	5.25E-02	2.76E+00	2.87E-01

Table A6
NEON Program Satellite Mission(s)
NIEL Equivalent Fluences in Silicon at the Center of Solid Aluminum Spheres
CL = 50%

Aluminum Solid-Sphere Shield Thickness			1 MeV Equivalent Proton Fluence #/cm ²	10 MeV Equivalent Proton Fluence #/cm ²	50 MeV Equivalent Proton Fluence #/cm ²	63 MeV Equivalent Proton Fluence #/cm ²	1 MeV Equivalent Neutron Fluence #/cm ²
mils	mm	g/cm ²					
1	0.0254	0.00686	6.731E+11	5.741E+12	1.168E+13	1.344E+13	1.857E+13
3	0.0762	0.02057	1.308E+11	1.115E+12	2.268E+12	2.611E+12	3.608E+12
10	0.2540	0.06855	2.630E+10	2.243E+11	4.562E+11	5.252E+11	7.257E+11
25	0.6350	0.17138	8.945E+09	7.630E+10	1.552E+11	1.786E+11	2.468E+11
50	1.2700	0.34276	4.502E+09	3.840E+10	7.809E+10	8.991E+10	1.242E+11
60	1.5240	0.41131	3.834E+09	3.270E+10	6.649E+10	7.656E+10	1.058E+11
75	1.9050	0.51414	3.195E+09	2.725E+10	5.541E+10	6.380E+10	8.815E+10
100	2.5400	0.68552	2.571E+09	2.193E+10	4.459E+10	5.134E+10	7.093E+10
125	3.1750	0.85690	2.155E+09	1.838E+10	3.737E+10	4.303E+10	5.945E+10
150	3.8100	1.02828	1.917E+09	1.635E+10	3.325E+10	3.828E+10	5.289E+10
200	5.0800	1.37104	1.709E+09	1.458E+10	2.964E+10	3.412E+10	4.715E+10
250	6.3500	1.71380	1.431E+09	1.221E+10	2.482E+10	2.858E+10	3.948E+10
300	7.6200	2.05656	1.278E+09	1.090E+10	2.216E+10	2.552E+10	3.526E+10
350	8.8900	2.39932	1.144E+09	9.759E+09	1.985E+10	2.285E+10	3.157E+10
400	10.1600	2.74208	1.065E+09	9.087E+09	1.848E+10	2.128E+10	2.940E+10
500	12.7000	3.42760	9.465E+08	8.074E+09	1.642E+10	1.890E+10	2.612E+10
600	15.2400	4.11312	8.514E+08	7.262E+09	1.477E+10	1.700E+10	2.349E+10
700	17.7800	4.79864	7.682E+08	6.553E+09	1.332E+10	1.534E+10	2.120E+10
800	20.3200	5.48416	6.895E+08	5.881E+09	1.196E+10	1.377E+10	1.902E+10
900	22.8600	6.16969	6.434E+08	5.488E+09	1.116E+10	1.285E+10	1.775E+10
1000	25.4000	6.85521	6.077E+08	5.184E+09	1.054E+10	1.214E+10	1.677E+10

Table A7
NEON Program Satellite Mission(s)
NIEL Equivalent Fluences in Silicon at the Center of Solid Aluminum Spheres
CL = 95%

Aluminum Solid-Sphere Shield Thickness			1 MeV Equivalent Proton Fluence #/cm ²	10 MeV Equivalent Proton Fluence #/cm ²	50 MeV Equivalent Proton Fluence #/cm ²	63 MeV Equivalent Proton Fluence #/cm ²	1 MeV Equivalent Neutron Fluence #/cm ²
mils	mm	g/cm ²					
1	0.0254	0.00686	1.366E+12	1.165E+13	2.369E+13	2.727E+13	3.768E+13
3	0.0762	0.02057	3.388E+11	2.890E+12	5.876E+12	6.766E+12	9.348E+12
10	0.2540	0.06855	8.737E+10	7.452E+11	1.515E+12	1.745E+12	2.411E+12
25	0.6350	0.17138	3.105E+10	2.649E+11	5.387E+11	6.202E+11	8.569E+11
50	1.2700	0.34276	1.407E+10	1.200E+11	2.441E+11	2.810E+11	3.883E+11
60	1.5240	0.41131	1.146E+10	9.772E+10	1.987E+11	2.288E+11	3.161E+11
75	1.9050	0.51414	9.019E+09	7.693E+10	1.564E+11	1.801E+11	2.489E+11
100	2.5400	0.68552	6.731E+09	5.741E+10	1.168E+11	1.344E+11	1.857E+11
125	3.1750	0.85690	5.334E+09	4.550E+10	9.253E+10	1.065E+11	1.472E+11
150	3.8100	1.02828	4.621E+09	3.942E+10	8.015E+10	9.228E+10	1.275E+11
200	5.0800	1.37104	3.848E+09	3.283E+10	6.675E+10	7.685E+10	1.062E+11
250	6.3500	1.71380	3.150E+09	2.687E+10	5.464E+10	6.291E+10	8.692E+10
300	7.6200	2.05656	2.764E+09	2.357E+10	4.794E+10	5.519E+10	7.626E+10
350	8.8900	2.39932	2.437E+09	2.079E+10	4.227E+10	4.866E+10	6.724E+10
400	10.1600	2.74208	2.244E+09	1.914E+10	3.892E+10	4.481E+10	6.191E+10
500	12.7000	3.42760	1.932E+09	1.648E+10	3.351E+10	3.858E+10	5.330E+10
600	15.2400	4.11312	1.724E+09	1.470E+10	2.990E+10	3.442E+10	4.756E+10
700	17.7800	4.79864	1.545E+09	1.318E+10	2.680E+10	3.086E+10	4.264E+10
800	20.3200	5.48416	1.379E+09	1.176E+10	2.392E+10	2.754E+10	3.805E+10
900	22.8600	6.16969	1.285E+09	1.096E+10	2.229E+10	2.567E+10	3.546E+10
1000	25.4000	6.85521	1.208E+09	1.030E+10	2.095E+10	2.412E+10	3.333E+10

Table A8
NEON Program Satellite Mission(s)
NIEL Equivalent Fluences in Gallium Arsenide at the Center of Solid Aluminum Spheres
CL = 50%

Aluminum Solid-Sphere Shield Thickness			1 MeV Equivalent Proton Fluence #/cm ²	10 MeV Equivalent Proton Fluence #/cm ²	50 MeV Equivalent Proton Fluence #/cm ²	63 MeV Equivalent Proton Fluence #/cm ²	1 MeV Equivalent Neutron Fluence (Ga) #/cm ²
mils	mm	g/cm ²					
1	0.0254	0.00686	5.130E+11	4.203E+12	7.387E+12	7.539E+12	3.016E+13
3	0.0762	0.02057	1.037E+11	8.498E+11	1.493E+12	1.524E+12	6.097E+12
10	0.2540	0.06855	2.222E+10	1.821E+11	3.200E+11	3.266E+11	1.307E+12
25	0.6350	0.17138	8.481E+09	6.950E+10	1.221E+11	1.247E+11	4.987E+11
50	1.2700	0.34276	4.759E+09	3.900E+10	6.853E+10	6.995E+10	2.798E+11
60	1.5240	0.41131	4.167E+09	3.414E+10	6.000E+10	6.124E+10	2.450E+11
75	1.9050	0.51414	3.593E+09	2.944E+10	5.173E+10	5.280E+10	2.112E+11
100	2.5400	0.68552	3.019E+09	2.473E+10	4.347E+10	4.437E+10	1.775E+11
125	3.1750	0.85690	2.630E+09	2.155E+10	3.787E+10	3.865E+10	1.546E+11
150	3.8100	1.02828	2.407E+09	1.973E+10	3.467E+10	3.538E+10	1.415E+11
200	5.0800	1.37104	2.185E+09	1.791E+10	3.147E+10	3.212E+10	1.285E+11
250	6.3500	1.71380	1.907E+09	1.563E+10	2.747E+10	2.803E+10	1.121E+11
300	7.6200	2.05656	1.752E+09	1.436E+10	2.523E+10	2.575E+10	1.030E+11
350	8.8900	2.39932	1.613E+09	1.322E+10	2.323E+10	2.371E+10	9.483E+10
400	10.1600	2.74208	1.522E+09	1.247E+10	2.192E+10	2.237E+10	8.950E+10
500	12.7000	3.42760	1.385E+09	1.135E+10	1.995E+10	2.036E+10	8.144E+10
600	15.2400	4.11312	1.272E+09	1.042E+10	1.832E+10	1.870E+10	7.480E+10
700	17.7800	4.79864	1.174E+09	9.621E+09	1.691E+10	1.726E+10	6.903E+10
800	20.3200	5.48416	1.074E+09	8.801E+09	1.547E+10	1.579E+10	6.315E+10
900	22.8600	6.16969	1.015E+09	8.316E+09	1.461E+10	1.492E+10	5.967E+10
1000	25.4000	6.85521	9.611E+08	7.876E+09	1.384E+10	1.413E+10	5.651E+10

Table A9
NEON Program Satellite Mission(s)
NIEL Equivalent Fluences in Gallium Arsenide at the Center of Solid Aluminum Spheres
CL = 95%

Aluminum Solid-Sphere Shield Thickness			1 MeV Equivalent Proton Fluence #/cm ²	10 MeV Equivalent Proton Fluence #/cm ²	50 MeV Equivalent Proton Fluence #/cm ²	63 MeV Equivalent Proton Fluence #/cm ²	1 MeV Equivalent Neutron Fluence (Ga) #/cm ²
mils	mm	g/cm ²					
1	0.0254	0.00686	1.028E+12	8.422E+12	1.480E+13	1.511E+13	6.043E+13
3	0.0762	0.02057	2.537E+11	2.079E+12	3.653E+12	3.729E+12	1.492E+13
10	0.2540	0.06855	6.574E+10	5.387E+11	9.467E+11	9.662E+11	3.865E+12
25	0.6350	0.17138	2.593E+10	2.124E+11	3.733E+11	3.811E+11	1.524E+12
50	1.2700	0.34276	1.350E+10	1.106E+11	1.944E+11	1.984E+11	7.937E+11
60	1.5240	0.41131	1.143E+10	9.363E+10	1.645E+11	1.679E+11	6.718E+11
75	1.9050	0.51414	9.407E+09	7.709E+10	1.355E+11	1.383E+11	5.531E+11
100	2.5400	0.68552	7.463E+09	6.115E+10	1.075E+11	1.097E+11	4.388E+11
125	3.1750	0.85690	6.241E+09	5.114E+10	8.987E+10	9.173E+10	3.669E+11
150	3.8100	1.02828	5.574E+09	4.568E+10	8.027E+10	8.193E+10	3.277E+11
200	5.0800	1.37104	4.815E+09	3.945E+10	6.933E+10	7.077E+10	2.831E+11
250	6.3500	1.71380	4.148E+09	3.399E+10	5.973E+10	6.097E+10	2.439E+11
300	7.6200	2.05656	3.741E+09	3.065E+10	5.387E+10	5.498E+10	2.199E+11
350	8.8900	2.39932	3.407E+09	2.792E+10	4.907E+10	5.008E+10	2.003E+11
400	10.1600	2.74208	3.185E+09	2.610E+10	4.587E+10	4.682E+10	1.873E+11
500	12.7000	3.42760	2.852E+09	2.337E+10	4.107E+10	4.192E+10	1.677E+11
600	15.2400	4.11312	2.593E+09	2.124E+10	3.733E+10	3.811E+10	1.524E+11
700	17.7800	4.79864	2.389E+09	1.958E+10	3.440E+10	3.511E+10	1.405E+11
800	20.3200	5.48416	2.185E+09	1.791E+10	3.147E+10	3.212E+10	1.285E+11
900	22.8600	6.16969	2.056E+09	1.684E+10	2.960E+10	3.021E+10	1.209E+11
1000	25.4000	6.85521	1.963E+09	1.608E+10	2.827E+10	2.885E+10	1.154E+11

Table A10
NEON Program Satellite Mission(s)
Integral LET Spectra for Galactic Cosmic Ray Ions (Z=1-92)
100 mils Aluminum Shielding

LET (MeV- cm ² /mg):	Flux > LET (#/cm ² /day):	
	Solar Maximum	Solar Minimum
1.01E-01	3.78E+02	7.96E+02
1.15E-01	2.58E+02	5.97E+02
1.30E-01	2.11E+02	5.00E+02
1.48E-01	1.92E+02	4.45E+02
1.68E-01	1.77E+02	4.00E+02
1.91E-01	1.55E+02	3.50E+02
2.17E-01	1.44E+02	3.17E+02
2.47E-01	1.25E+02	2.76E+02
2.80E-01	1.06E+02	2.36E+02
3.18E-01	9.45E+01	2.08E+02
3.62E-01	7.07E+01	1.64E+02
4.11E-01	6.24E+01	1.43E+02
4.67E-01	5.52E+01	1.25E+02
5.30E-01	5.13E+01	1.13E+02
6.03E-01	4.75E+01	1.02E+02
6.85E-01	4.30E+01	9.07E+01
7.78E-01	3.89E+01	8.11E+01
8.84E-01	3.55E+01	7.29E+01
1.00E+00	3.17E+01	6.47E+01
1.14E+00	2.81E+01	5.72E+01
1.30E+00	9.43E+00	2.86E+01
1.47E+00	4.57E+00	1.87E+01
1.67E+00	2.81E+00	1.35E+01
1.90E+00	1.87E+00	1.00E+01
2.16E+00	1.28E+00	7.50E+00
2.45E+00	8.90E-01	5.62E+00
2.79E+00	6.29E-01	4.22E+00
3.16E+00	4.49E-01	3.17E+00
3.59E+00	3.24E-01	2.38E+00
4.08E+00	2.35E-01	1.78E+00
4.64E+00	1.71E-01	1.33E+00
5.27E+00	1.24E-01	9.84E-01
5.99E+00	9.08E-02	7.29E-01
6.80E+00	6.66E-02	5.39E-01
7.73E+00	4.88E-02	3.96E-01
8.78E+00	3.63E-02	2.96E-01
9.97E+00	2.68E-02	2.18E-01
1.13E+01	1.95E-02	1.59E-01
1.29E+01	1.39E-02	1.14E-01
1.46E+01	9.76E-03	7.94E-02
1.66E+01	6.85E-03	5.57E-02

Table A10 (continued)
NEON Program Satellite Mission(s)
Integral LET Spectra for Galactic Cosmic Ray Ions (Z=1-92)
100 mils Aluminum Shielding

LET (MeV- cm²/mg):	Flux > LET (#/cm²/day):	
	Solar Maximum	Solar Minimum
1.89E+01	4.64E-03	3.77E-02
2.14E+01	2.87E-03	2.33E-02
2.44E+01	1.52E-03	1.23E-02
2.77E+01	3.53E-04	2.85E-03
3.14E+01	5.12E-06	3.37E-05
3.57E+01	2.56E-06	1.65E-05
4.06E+01	1.65E-06	1.06E-05
4.61E+01	1.09E-06	6.95E-06
5.24E+01	7.02E-07	4.48E-06
5.95E+01	4.18E-07	2.64E-06
6.76E+01	2.27E-07	1.42E-06
7.68E+01	1.03E-07	6.45E-07
8.72E+01	1.96E-08	1.22E-07
9.91E+01	7.51E-10	4.71E-09

Table A11
NEON Program Satellite Mission(s)
Integral LET Spectra for the October 1989 Solar Particle Event (Z=1-92)
100 mils Aluminum Shielding

LET (MeV- cm ² /mg):	Flux > LET (#/cm ² /sec):	
	Worst 5 Minutes	Worst Day
1.01E-01	1.05E+03	2.78E+02
1.15E-01	7.94E+02	2.10E+02
1.30E-01	5.95E+02	1.57E+02
1.48E-01	4.43E+02	1.17E+02
1.68E-01	3.27E+02	8.64E+01
1.91E-01	2.37E+02	6.26E+01
2.17E-01	1.71E+02	4.52E+01
2.47E-01	1.22E+02	3.22E+01
2.80E-01	8.55E+01	2.26E+01
3.18E-01	5.92E+01	1.57E+01
3.62E-01	4.02E+01	1.06E+01
4.11E-01	2.59E+01	6.84E+00
4.67E-01	1.46E+01	3.87E+00
5.30E-01	5.76E+00	1.53E+00
6.03E-01	4.41E+00	1.17E+00
6.85E-01	3.36E+00	8.95E-01
7.78E-01	2.55E+00	6.78E-01
8.84E-01	1.91E+00	5.11E-01
1.00E+00	1.42E+00	3.80E-01
1.14E+00	1.03E+00	2.77E-01
1.30E+00	7.27E-01	1.96E-01
1.47E+00	4.71E-01	1.28E-01
1.67E+00	2.56E-01	7.04E-02
1.90E+00	2.16E-01	5.95E-02
2.16E+00	1.82E-01	5.00E-02
2.45E+00	1.52E-01	4.18E-02
2.79E+00	1.26E-01	3.48E-02
3.16E+00	1.05E-01	2.89E-02
3.59E+00	8.62E-02	2.38E-02
4.08E+00	7.06E-02	1.95E-02
4.64E+00	5.70E-02	1.58E-02
5.27E+00	4.48E-02	1.24E-02
5.99E+00	3.54E-02	9.81E-03
6.80E+00	2.72E-02	7.54E-03
7.73E+00	1.99E-02	5.53E-03
8.78E+00	1.55E-02	4.30E-03
9.97E+00	1.17E-02	3.24E-03
1.13E+01	8.71E-03	2.42E-03
1.29E+01	6.24E-03	1.73E-03
1.46E+01	4.23E-03	1.17E-03
1.66E+01	2.98E-03	8.30E-04

Table A11 (continued)
NEON Program Satellite Mission(s)
Integral LET Spectra for the October 1989 Solar Particle Event (Z=1-92)
100 mils Aluminum Shielding

LET (MeV- cm²/mg):	Flux > LET (#/cm²/sec):	
	Worst 5 Minutes	Worst Day
1.89E+01	2.10E-03	5.83E-04
2.14E+01	1.36E-03	3.78E-04
2.44E+01	7.97E-04	2.22E-04
2.77E+01	2.11E-04	5.86E-05
3.14E+01	7.99E-07	2.22E-07
3.57E+01	2.69E-07	7.50E-08
4.06E+01	1.31E-07	3.65E-08
4.61E+01	7.99E-08	2.22E-08
5.24E+01	5.11E-08	1.42E-08
5.95E+01	2.57E-08	7.14E-09
6.76E+01	1.40E-08	3.90E-09
7.68E+01	7.89E-09	2.19E-09
8.72E+01	2.15E-09	5.99E-10
9.91E+01	3.91E-11	1.09E-11

Table A12
NEON Program Satellite Mission(s)
Worst-Case Integral Proton Fluxes for SEE Evaluation (5 Min. Avg.)
100 mils Aluminum Shielding

Energy, E (MeV):	CL 50% Flux > E (p/cm²/s):	CL 95% Flux > E (p/cm²/s):
1.00E-01	1.06E+05	7.57E+05
1.12E-01	1.06E+05	7.56E+05
1.26E-01	1.06E+05	7.56E+05
1.41E-01	1.06E+05	7.56E+05
1.58E-01	1.06E+05	7.56E+05
1.78E-01	1.06E+05	7.56E+05
2.00E-01	1.06E+05	7.56E+05
2.24E-01	1.06E+05	7.56E+05
2.51E-01	1.06E+05	7.56E+05
2.82E-01	1.06E+05	7.56E+05
3.16E-01	1.06E+05	7.56E+05
3.55E-01	1.06E+05	7.56E+05
3.98E-01	1.06E+05	7.56E+05
4.47E-01	1.06E+05	7.56E+05
5.01E-01	1.06E+05	7.55E+05
5.62E-01	1.06E+05	7.55E+05
6.31E-01	1.06E+05	7.55E+05
7.08E-01	1.06E+05	7.55E+05
7.94E-01	1.06E+05	7.54E+05
8.91E-01	1.06E+05	7.54E+05
1.00E+00	1.06E+05	7.53E+05
1.12E+00	1.06E+05	7.53E+05
1.26E+00	1.05E+05	7.52E+05
1.41E+00	1.05E+05	7.51E+05
1.58E+00	1.05E+05	7.50E+05
1.78E+00	1.05E+05	7.49E+05
2.00E+00	1.05E+05	7.47E+05
2.24E+00	1.04E+05	7.46E+05
2.51E+00	1.04E+05	7.44E+05
2.82E+00	1.03E+05	7.41E+05
3.16E+00	1.03E+05	7.38E+05
3.55E+00	1.02E+05	7.34E+05
3.98E+00	1.01E+05	7.29E+05
4.47E+00	9.98E+04	7.24E+05
5.01E+00	9.86E+04	7.17E+05
5.62E+00	9.70E+04	7.09E+05
6.31E+00	9.52E+04	6.99E+05
7.08E+00	9.30E+04	6.88E+05
7.94E+00	9.05E+04	6.75E+05
8.91E+00	8.76E+04	6.59E+05

Table A12 (continued)
NEON Program Satellite Mission(s)
Worst-Case Integral Proton Fluxes for SEE Evaluation (5 Min. Avg.)
100 mils Aluminum Shielding

Energy, E (MeV):	CL 50% Flux > E (p/cm²/s):	CL 95% Flux > E (p/cm²/s):
1.00E+01	8.42E+04	6.41E+05
1.12E+01	8.04E+04	6.20E+05
1.26E+01	7.61E+04	5.96E+05
1.41E+01	7.14E+04	5.69E+05
1.58E+01	6.62E+04	5.40E+05
1.78E+01	6.07E+04	5.07E+05
2.00E+01	5.51E+04	4.72E+05
2.24E+01	4.93E+04	4.35E+05
2.51E+01	4.37E+04	3.97E+05
2.82E+01	3.80E+04	3.58E+05
3.16E+01	3.27E+04	3.20E+05
3.55E+01	2.78E+04	2.82E+05
3.98E+01	2.33E+04	2.46E+05
4.47E+01	1.94E+04	2.12E+05
5.01E+01	1.59E+04	1.81E+05
5.62E+01	1.29E+04	1.53E+05
6.31E+01	1.03E+04	1.27E+05
7.08E+01	8.21E+03	1.05E+05
7.94E+01	6.45E+03	8.55E+04
8.91E+01	5.03E+03	6.89E+04
1.00E+02	3.93E+03	5.50E+04
1.12E+02	3.11E+03	4.38E+04
1.26E+02	2.44E+03	3.44E+04
1.41E+02	1.90E+03	2.67E+04
1.58E+02	1.47E+03	2.06E+04
1.78E+02	1.12E+03	1.58E+04
2.00E+02	8.59E+02	1.21E+04
2.24E+02	6.50E+02	9.15E+03
2.51E+02	4.87E+02	6.85E+03
2.82E+02	3.60E+02	5.06E+03
3.16E+02	2.63E+02	3.70E+03
3.55E+02	1.92E+02	2.70E+03
3.98E+02	1.38E+02	1.95E+03
4.47E+02	9.85E+01	1.39E+03